

Adaptive Approximation

Adaptive Finite Element Methods for Elliptic PDEs — Lecture 4

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Why adapt?

- **Limitation of uniform meshes:** On a quasi-uniform mesh the convergence rate is capped by the **global** L^2 -based Sobolev regularity of u .
- **Adaptive remedy:** Adaptivity instead **tailors the mesh to u** through a nonlinear process and measures the regularity in an L^p -based Sobolev space with $p < 2$.

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The plan of this lecture

- **Why:** error equidistribution — a 1-D graded-mesh model, then the 2-D heuristic; the singular class W_1^2 reaches the same rate as H^2 .
- **How:** *bisection* (newest-vertex bisection) builds graded, conforming, shape-regular meshes by purely local operations.
- **Cost:** the Binev–Dahmen– DeVore / Stevenson complexity bound $\#\mathcal{T}_k - \#\mathcal{T}_0 \lesssim \sum_j \#\mathcal{M}_j$.
- **Optimality:** a *greedy* algorithm; W_p^2 regularity ($p > 1$) yields the optimal rate $N^{-1/2}$.

Nonlinear approximation: the mesh is designed for the particular u at hand.

Roadmap

- 1 Why adapt? Error equidistribution
- 2 The Bisection Method
- 3 Complexity of Bisection
- 4 The Greedy Algorithm
- 5 Summary

Why adapt? Error equidistribution

One-dimensional model problem: uniform mesh

- **Setting:** Let $\Omega = (0, 1)$ and approximate an absolutely continuous v by a piecewise-constant v_N on a mesh with N intervals $I_i = [x_{i-1}, x_i]$.

- **Uniform mesh - smooth data:**

If $v \in W_\infty^1(\Omega)$ and $\mathcal{T}$ is uniform ($h_i = h = 1/N$), integrating $|v'|$ over I_i gives

$$|v(x) - v_N(x)| \leq h \|v'\|_{L^\infty(\Omega)} \quad \Rightarrow \quad \|v - v_N\|_{L^\infty(\Omega)} \leq \frac{1}{N} |v|_{W_\infty^1(\Omega)}.$$

- **Remarks:**

- The rate N^{-1} requires the **strong** regularity $v \in W_\infty^1$.
- What if v' is unbounded (a singularity)? A uniform mesh no longer delivers this rate.
- If $v \in C^{0,\gamma}$ for $0 < \gamma < 1$, then $\|v - v_N\|_{L^\infty(\Omega)} \lesssim N^{-\gamma}$.

One-dimensional model problem: Graded mesh

- **Rough function:** Let $v \in W_1^1(\Omega)$ only, possibly with unbounded derivative. Introduce the monotone **error-accumulation function**

$$\phi(x) = \int_0^x |v'(s)| \, ds, \quad \phi(0) = 0, \quad \phi(1) = |v|_{W_1^1(\Omega)}.$$

- **Grade by equidistributing the *range* of ϕ :**

Choose nodes x_i so that

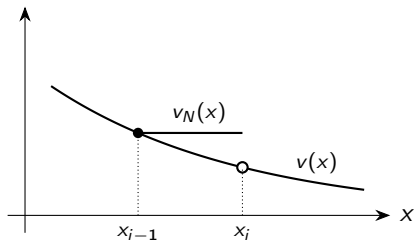
$$\phi(x_i) = \frac{i}{N} |v|_{W_1^1(\Omega)} \quad (0 \leq i \leq N),$$

i.e. cut the range of ϕ into N equal pieces.

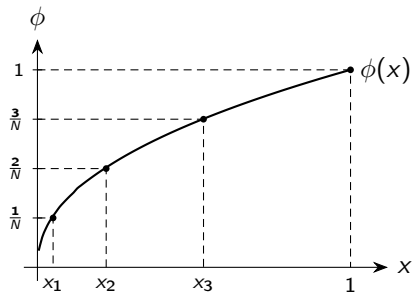
- **Error equidistribution:** For $x \in I_i$ this **equalizes** the local errors:

$$|v(x) - v_N(x)| \leq \int_{x_{i-1}}^{x_i} |v'| \, ds = \phi(x_i) - \phi(x_{i-1}) = \frac{1}{N} |v|_{W_1^1(\Omega)}, \quad \Rightarrow \quad \|v - v_N\|_{L^\infty(\Omega)} \leq \frac{1}{N} |v|_{W_1^1(\Omega)}.$$

The 1-D mechanism for error equidistribution



(a) piecewise-constant v_N



(b) equidistributing the range of ϕ

Nodes x_i (via $\phi(x_i) = i/N$) make every subinterval carry the same error $1/N$; they **cluster where $|v'|$ is large**.

The singular example $v(x) = x^\gamma$

- **A singular test function:** Take $v(x) = x^\gamma$, $0 < \gamma < 1$. Then

$$v'(x) = \gamma x^{\gamma-1} \notin L^\infty(0, 1),$$

so uniform mesh gives $\|v - v_N\|_{L^\infty(\Omega)} \lesssim h^\gamma$; yet $v \in W_1^1(\Omega)$ and

$$\phi(x) = \int_0^x \gamma s^{\gamma-1} ds = x^\gamma = v(x).$$

- **Error accumulation:** The error-accumulation function **is the solution itself**: its singular behavior at the origin is exactly what makes a graded mesh profitable.
- **Sobolev numbers:** The graded mesh recovers the *same* $O(1/N)$ rate for the rougher class W_1^1 that the uniform mesh gives for W_∞^1 . This **hinges on equality of the Sobolev numbers**:

$$\text{sob}(W_1^1) = 1 - \frac{1}{1} \geq 0 - \frac{1}{\infty} = \text{sob}(L^\infty).$$

Heuristics in two dimensions

- **Local interpolation estimate:** The starting point is a local interpolation estimate carrying a **vanishing power of h_T** . One has $W_1^2(\Omega) \hookrightarrow C^0(\bar{\Omega})$, so $I_{\mathcal{T}} u$ is well defined and

$$\|\nabla(u - I_{\mathcal{T}} u)\|_{L^2(T)} \leq C |u|_{W_1^2(T)} =: e_{\mathcal{T}}(u, T) \quad \forall T \in \mathcal{T}.$$

- **Why the absence of h_T matters?** The estimate is written with equal Sobolev numbers

$$\text{sob}(W_1^2) = 2 - \frac{2}{1} = 0 = 1 - \frac{2}{2} = \text{sob}(H^1),$$

whence **no positive power of h_T** . This scale invariance makes the discrete optimization below clean.

- **Space $W_p^2(\Omega)$:** results below for $W_1^2(\Omega)$ are valid for $W_p^2(\Omega)$ with $p > 1$, because $W_p^2(\Omega) \hookrightarrow W_1^2(\Omega)$.

Discrete minimization problem

- **Total error bound:** Collect the surrogate local errors into $\mathbf{e} = (e_{\mathcal{T}}(u, T))_{T \in \mathcal{T}} \in \mathbb{R}^N$, $N = \#\mathcal{T}$, and consider the total (squared) error which **accumulates in ℓ^2**

$$E_{\mathcal{T}}(u)^2 := \sum_{T \in \mathcal{T}} e_{\mathcal{T}}(u, T)^2.$$

- **L^1 -constraint:** Accumulating the surrogate errors in ℓ^1 leads to the global regularity $W_1^2(\Omega)$.

$$\sum_{T \in \mathcal{T}} e_{\mathcal{T}}(u, T) = C \|u\|_{W_1^2(\Omega)}.$$

The **ℓ^1 -summability** reflects regularity $W_1^2(\Omega)$, i.e. the equality $\text{sob}(W_1^2) = \text{sob}(H^1)$.

- **Question:** Among all ways of distributing the fixed quantity $\sum_{T \in \mathcal{T}} e_{\mathcal{T}}(u, T)$ over N elements, which minimizes $E_{\mathcal{T}}(u)^2$?

Equidistribution is optimal

- **Lagrangian:** Idealize $e_{\mathcal{T}}(u, T)$ to range over all nonnegative reals and form the Lagrangian

$$\mathcal{L}[e, \lambda] = \sum_{T \in \mathcal{T}} e_{\mathcal{T}}(u, T)^2 - \lambda \left(\sum_{T \in \mathcal{T}} e_{\mathcal{T}}(u, T) - C \|u\|_{W_1^2(\Omega)} \right).$$

- **Optimality condition:** The condition $\partial \mathcal{L} / \partial e_{\mathcal{T}}(u, T) = 2 e_{\mathcal{T}}(u, T) - \lambda = 0$ forces

$$e_{\mathcal{T}}(u, T) = \frac{\lambda}{2} \quad \forall T \in \mathcal{T} :$$

the local error is the **same on every element**.

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the local error is the **same on every element**.

- **Conclusion:**

A graded mesh is quasi-optimal *if the local error is equidistributed*.

The resulting rate

- **Eliminating λ :** Equidistribution $e_{\mathcal{T}}(u, T) = \frac{\lambda}{2}$ gives

$$E_{\mathcal{T}}(u)^2 = \frac{\lambda^2}{4} N, \quad C |u|_{W_1^2(\Omega)} = \frac{\lambda}{2} N.$$

Eliminating λ yields the optimal decay rate

$$\|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(\Omega)} \leq C |u|_{W_1^2(\Omega)} N^{-1/2} \leq C |\Omega|^{1/q} |u|_{W_p^2(\Omega)} N^{-1/2},$$

where $q = \frac{p}{p-1}$ for $p > 1$.

- **Interpretation:**

- **Optimal rate:** Same rate $N^{-1/2}$ as *uniform* refinement for $u \in H^2(\Omega)$;
- **Weaker regularity:** It requires only $u \in W_p^2(\Omega)$, with $p \geq 1$, a **much larger space**.
- **Nonlinear approximation:** $\mathcal{T}$ is designed for the given u .

Example: W_p^2 -regularity at a reentrant corner

- Corner singularity:** Let $u(r, \theta) \sim r^\gamma$, $0 < \gamma < 1$ — the typical corner singularity. Then $D^2 u \sim r^{\gamma-2} \notin L^2$, so $u \notin H^2(\Omega)$; but

$$\|D^2 u\|_{L^p(\Omega)}^p = c \int_0^1 r^{p(\gamma-2)} r \, dr = c \int_0^1 r^{p(\gamma-2)+1} \, dr < \infty \implies u \in W_p^2(\Omega),$$

provided $1 \leq p < p_0 = \frac{2}{2-\gamma}$.

- Optimal rate via grading:** Hence $|u|_{W_p^2(\Omega)} < \infty$, and a mesh that equidistributes the local energy error — necessarily **graded toward the corner** — achieves

$$\|\nabla(u - I_{\mathcal{T}} u)\|_{L^2(\Omega)} \lesssim |u|_{W_p^2(\Omega)} N^{-1/2},$$

the optimal 2-D rate *despite* $u \notin H^2(\Omega)$.

Fractional regularity and the general rate

- **Fractional smoothness:** What dictates the exponent? For piecewise linears ($k = 1$) approximation theory allows fractional $1 < s \leq 2$. The local estimate persists,

$$\|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(T)} \leq C |u|_{W_p^s(\omega_T)} = e_{\mathcal{T}}(u, T) \quad \forall T \in \mathcal{T}, \quad \text{sob}(W_p^s) - \text{sob}(H^1) = s - \frac{2}{p} = 0,$$

with $I_{\mathcal{T}}$ a quasi-interpolant and ω_T a discrete neighborhood.

- **Accumulating in ℓ^p and ℓ^2 :** $e_{\mathcal{T}}(u, T) = \Lambda$ (constant) yields

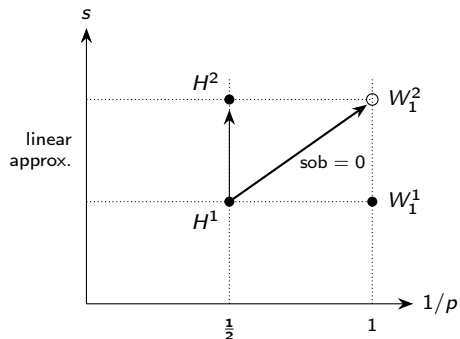
$$\sum_{T \in \mathcal{T}} e_{\mathcal{T}}(u, T)^p = \Lambda^p N \approx |u|_{W_p^s(\Omega)}^p, \quad E_{\mathcal{T}}(u)^2 = \Lambda^2 N.$$

- **Outcome:** $\Lambda \leq C |u|_{W_p^s(\Omega)} N^{-1/p}$, and altogether

$$\|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(\Omega)} \leq C |u|_{W_p^s(\Omega)} N^{-\frac{s-1}{2}}.$$

The numerator $s - 1$ is the difference of differentiability indices between W_p^s and H^1 .

The DeVore diagram



The oblique Sobolev line $\text{sob}(W_p^s) = s - \frac{2}{p} = 0$ (that of H^1) governs nonlinear approximation.

- **Linear** theory raises s from 1 to 2 with $p = 2$: reaches H^2 .
- **Nonlinear** theory trades integrability ($p = 2 \rightarrow 1$) for differentiability ($s = 1 \rightarrow 2$): reaches W_1^2 .

Both attain the optimal rate $N^{-(s-1)/2}$: graded meshes with regularity W_p^s while uniform meshes with regularity H^s .

Practical question:

- How to **construct** graded meshes automatically and locally that equidistribute the error, while preserving conformity and shape regularity?
- **The answer is bisection.**

The Bisection Method

Newest-vertex bisection

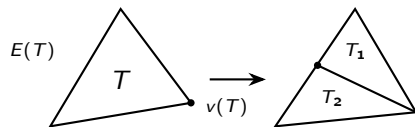
- **The refinement engine:** The refinement engine rests on a single operation: **bisection** of one element. Each T carries a distinguished **refinement edge** $E(T)$ and opposite a vertex $v(T)$.

Bisecting T joins $v(T)$ to the midpoint of $E(T)$, producing children T_1, T_2 sharing the newly created vertex $v(T_1) = v(T_2)$; each child gets its own refinement edge opposite the new vertex.

The *bisection method* involves two ingredients:

- a **labeling** of $\mathcal{T}_0$ fixing each element's refinement edge;
- a **rule** propagating labels to the children.

For $d = 2$ this is **newest-vertex bisection**; for $d > 2$ one adds element *type* and *vertex order*.



$T \rightarrow T_1, T_2$; $v(T_1) = v(T_2)$ is the new vertex.

Generation and labeling

- Generation:** The **generation** $g(T)$ is the number of bisections needed to create T from its ancestor $T_0 \in \mathcal{T}_0$. Since each bisection halves area and $h_T = |T|^{1/2}$,

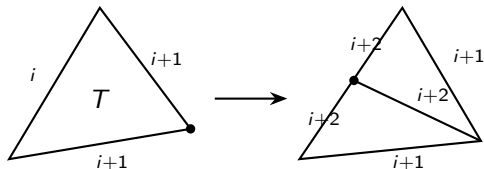
$$h_T = 2^{-g(T)/2} h_{T_0}, \quad T_0 \in \mathcal{T}_0 \text{ the ancestor of } T.$$

- Labeling:** Assign the ordered edge label $(i, i+1, i+1)$ to each T of generation $g(T) = i$, the level- i edge being $E(T)$.

Bisection splits the level- i edge and labels both new half-edges *and* the bisector $i+2$, leaving the rest unchanged:

$$(i, i+1, i+1) \longrightarrow (i+1, i+2, i+2)$$

for both children; the level- $(i+1)$ edges are the new refinement edges.



Shape regularity

- **Bisection preserves shape:** *bisection never degrades element shape*. The labeling prevents consecutive bisections of the same edge, inducing a rotation of refinement edges – a key geometric property.

Lemma 1 (shape regularity)

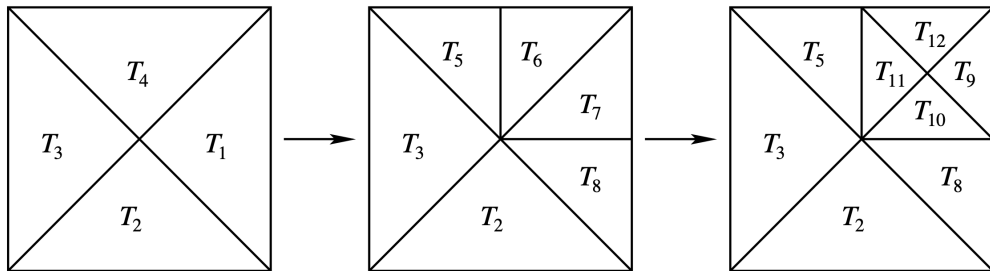
*Every partition generated from $\mathcal{T}_0$ by newest-vertex bisection satisfies a uniform minimal-angle condition: the ratio of element diameter to inscribed-ball diameter is bounded by a constant depending only on $\mathcal{T}_0$. Equivalently, the descendants of each $T_0 \in \mathcal{T}_0$ fall into *finitely many similarity classes*.*

- **The refinement order:** Write $\mathbb{T}$ for the conforming refinements of $\mathcal{T}_0$ reachable by bisection, and $\mathcal{T}_* \geq \mathcal{T}$ when $\mathcal{T}_*$ refines $\mathcal{T}$. This partial order $(\mathbb{T}, \geq)$ is the arena of the theory: the loop (SEMR) **SOLVE** $\rightarrow$ **ESTIMATE** $\rightarrow$ **MARK** $\rightarrow$ **REFINE**, discussed in Lecture 6, produces

$$\mathcal{T}_0 \leq \mathcal{T}_1 \leq \cdots \leq \mathcal{T}_k \leq \cdots,$$

and shape regularity keeps all interpolation and a-posteriori constants *uniform* along the way.

A sequence of bisection meshes



Meshes $\{\mathcal{T}_k\}_{k=0}^2$ from $\mathcal{T}_0 = \{T_i\}_{i=1}^4$ with longest edges labeled for bisection.
 $\mathcal{T}_1$ arises by bisecting T_1, T_4 ; $\mathcal{T}_2$ by refining T_6, T_7 . Every intermediate mesh is conforming.

Compatible labeling

- Compatible initial labeling:** For refinement to terminate and stay conforming, an element and its neighbor must **agree** on the shared refinement edge. In 2-D this is secured by a *compatible initial labeling* of $\mathcal{T}_0$: label every edge 0 or 1 so that

every $T \in \mathcal{T}_0$ has exactly two edges labeled 1 and one labeled 0,

the label-0 edge being its refinement edge (the generation-0 instance of $(i, i+1, i+1)$).

Compatible patch

If two elements share their refinement edge they form a **compatible patch**. When bisected *together*, inserting one vertex and creating *no hanging node*, the refinement remains local to the patch.

- Always arrangeable:** Such a labeling is not automatic but always arrangeable after refining $\mathcal{T}_0$: For $d = 2$ bisect every element of $\mathcal{T}_0$ twice and label the two newest edges 0, the rest 1. For $d > 2$ the analogous type/vertex-order condition is more delicate but still available.

Complexity of Bisection

The REFINE module and completion chains

- Module REFINE:** Given $\mathcal{T} \in \mathbb{T}$ and a marked set $\mathcal{M} \subset \mathcal{T}$, the module $\mathcal{T}_* = \text{REFINE}(\mathcal{T}, \mathcal{M})$ returns the **smallest conforming refinement** in which every $T \in \mathcal{M}$ has been bisected at least once.
- Hanging node:** Bisecting a marked T across $E(T)$ generally leaves a **hanging node** on the neighbor across that edge.
- Restoring conformity:** Conformity is restored by bisecting the neighbor too — which may create a new hanging node, and so on.
- Completion chain:** The completion runs along a **chain** of elements linked by refinement edges, each bisected in turn until a compatible patch is reached (terminating at $\partial\Omega$ or a compatible pair). Crucially, the chain can be as long as the generation $g(T)$ of the marked element.

Recursive refinement is nonlocal

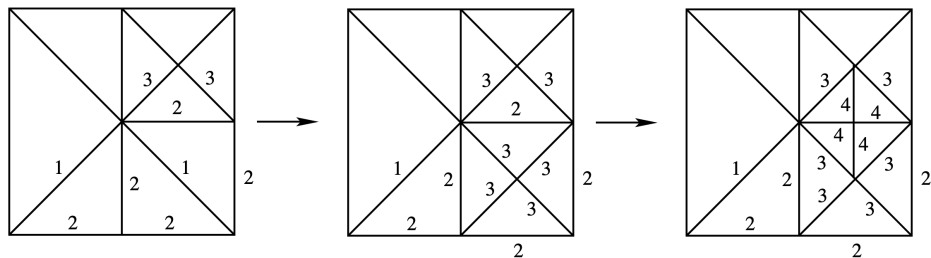


Figure: Refinement chain: Refining T_{10} forces the chain $\{T_{10}, T_8, T_2\}$, resolved in reverse: first T_2 (refinement edge on $\partial\Omega$, a compatible patch on its own), then T_8 , finally T_{10} . The chain length is $g(T_{10}) = 3$, and it **grows without bound** as the mesh is refined.

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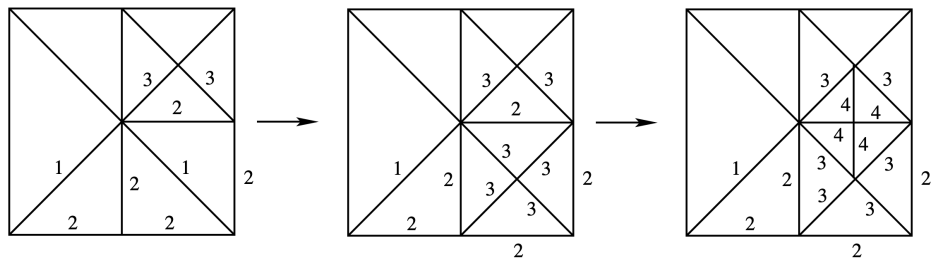


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Question: How can we related the cardinality $\#\mathcal{M}$ of $\mathcal{M}$ with $\#\mathcal{T}_*$?

Complexity of REFINE: a naive bound

- **Nonlocal process:** Because one mark can trigger a completion chain, the tempting *local* bound

$$\#\mathcal{T}_* - \#\mathcal{T} \leq \Lambda_0 \#\mathcal{M}$$

does not hold with a Λ_0 independent of the refinement level: the local cost may be as large as $g(T)$ for $T \in \mathcal{M}$, which grows without bound.

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- **The remedy:**

Do not bound a single step but the **cumulative** cost over the whole refinement history. The completion process is in average local and the **total** number of marks controls it.

- **Delicate property:** This is due to **Binev–Dahmen– DeVore** ($d = 2$) and **Stevenson** ($d \geq 3$).

Complexity of REFINE: Rigorous bound

Theorem 1 (complexity of bisection)

Suppose $\mathcal{T}_0$ satisfies the compatible labeling for $d = 2$ (the analogous type/vertex-order condition of Stevenson for $d > 2$). Let $\mathcal{M}_j \subset \mathcal{T}_j$ be the marked set at step j and $\mathcal{T}_{j+1} = \text{REFINE}(\mathcal{T}_j, \mathcal{M}_j)$. Then there is a constant Λ_0 , depending only on $\mathcal{T}_0$ and d , such that for all $k \geq 1$

$$\#\mathcal{T}_k - \#\mathcal{T}_0 \leq \Lambda_0 \sum_{j=0}^{k-1} \#\mathcal{M}_j.$$

If each marked element is bisected a fixed number $b \geq 1$ of times (e.g. $b = 2$), the bound persists with Λ_0 also depending on b .

- **Key idea:** The above bound relies on the construction of an allocation function $\lambda(T, T_*)$ with suitable properties, as discussed next.

Heuristics: Allocation Function

- Each element $T_* \in \mathcal{M}$ is assigned a fixed amount C_1 of money to spend on refined elements in $\mathcal{T}$.
- Let $\lambda(T, T_*)$ be the portion of money spent by T_* on T . Then it must hold

$$\sum_{T \in \mathcal{T} \setminus \mathcal{T}_0} \lambda(T, T_*) \leq C_1 \quad \forall T_* \in \mathcal{M} = \cup_{j=0}^{k-1} \mathcal{M}_j.$$

- The investment of all elements in $\mathcal{M}$ is fair in the sense that each $T \in \mathcal{T} \setminus \mathcal{T}_0$ gets at least a fixed amount C_2 , whence

$$\sum_{T_* \in \mathcal{M}} \lambda(T, T_*) \geq C_2 \quad \forall T \in \mathcal{T} \setminus \mathcal{T}_0.$$

- Summing up yields

$$C_2(\#\mathcal{T} - \#\mathcal{T}_0) \leq \sum_{T \in \mathcal{T} \setminus \mathcal{T}_0} \sum_{T_* \in \mathcal{M}} \lambda(T, T_*) = \sum_{T_* \in \mathcal{M}} \sum_{T \in \mathcal{T} \setminus \mathcal{T}_0} \lambda(T, T_*) \leq C_1 \#\mathcal{M}.$$

Why the bound matters

Combinatorial backbone of optimality:

The total number of elements $\#\mathcal{T}_k - \#\mathcal{T}_0$ created by REFINE at step k is controlled by the total number of marked elements $\sum_{j=0}^{k-1} \#\mathcal{M}_j$ from step 1 to step k :

$$\#\mathcal{T}_k - \#\mathcal{T}_0 \leq \Lambda_0 \sum_{j=0}^{k-1} \#\mathcal{M}_j.$$

- **The stakes:** Without it, the completion process — however local it looks element by element — could in principle overwhelm the savings of adaptivity.
- **The payoff:** With it, the cost of keeping meshes conforming is provably a **fixed multiple** of the useful refined elements $\sum_{j=0}^{k-1} \#\mathcal{M}_j$.
- **Looking ahead:** This is exactly the ingredient the optimality theory (Lecture 9) needs.

The Greedy Algorithm

Greedy adaptive algorithm

- Goal:** Given $u \in H_0^1(\Omega) \cap W_p^2(\Omega)$, $p > 1$, construct a near-optimal mesh by iterative bisection. Define the **local energy error**

$$e_{\mathcal{T}}(T) = e_{\mathcal{T}}(u, T) := \|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(T)},$$

and let $\delta > 0$ be an error tolerance.

GREEDY $(\mathcal{T}_0, \delta)$

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function GREEDY( $\mathcal{T}_0, \delta$ )
 $\mathcal{T} := \mathcal{T}_0$ 
while  $\mathcal{M} := \{T \in \mathcal{T} : e_{\mathcal{T}}(T) > \delta\} \neq \emptyset$ :
     $\mathcal{T} \leftarrow \text{REFINE}(\mathcal{T}, \mathcal{M})$ 
return  $\mathcal{T}$ 
  
```

This assumes we know u and can evaluate $e_{\mathcal{T}}(T)$: a **theoretical benchmark** for what adaptive methods can achieve. In practice $e_{\mathcal{T}}(T)$ is replaced by a computable estimator (Lecture 5).

Optimal greedy approximation

Theorem 2 (optimal greedy approximation)

If $u \in H_0^1(\Omega) \cap W_p^2(\Omega)$ for some $p > 1$ and $d = 2$, the output $\mathcal{T}$ of $\text{GREEDY}(\mathcal{T}_0, \delta)$ satisfies

- ❶ $\|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(\Omega)} \leq \delta (\#\mathcal{T})^{1/2};$
- ❷ $\#\mathcal{T} - \#\mathcal{T}_0 \lesssim \delta^{-1} |\Omega|^{1-1/p} |u|_{W_p^2(\Omega)}.$

In particular, with $N = \#\mathcal{T} \geq 2\#\mathcal{T}_0$,

$$\|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(\Omega)} \lesssim |\Omega|^{1-1/p} |u|_{W_p^2(\Omega)} N^{-1/2}.$$

- **Interpretation:** $N^{-1/2}$ is the optimal rate for $d = 2$ but it is achieved with regularity $W_p^2(\Omega)$ with $p > 1$ rather than the limiting space $W_1^2(\Omega)$. This is due to the completion process of REFINE to keep conformity. We prove this theorem in five steps.

Proof — Steps 1 & 2

- Step 1 — Local error bound (termination + statement (1)):** By the local interpolation estimate $e_{\mathcal{T}}(T) \leq C h_T^r |u|_{W_p^2(T)}$ with

$$r = \text{sob}(W_p^2) - \text{sob}(H^1) = 2(1 - 1/p) > 0,$$

refining strictly decreases $e_{\mathcal{T}}(T)$ and the algorithm **terminates**. At termination $e_{\mathcal{T}}(T) \leq \delta$ for all $T \in \mathcal{T}$, thereby giving

$$\|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(\Omega)}^2 = \sum_{T \in \mathcal{T}} e_{\mathcal{T}}(T)^2 \leq \delta^2 \#\mathcal{T},$$

which is statement (1).

- Step 2 — Counting elements:** Let $\mathcal{M}$ be the **total** set of all elements ever marked. By the complexity theorem $\#\mathcal{T} - \#\mathcal{T}_0 \leq \Lambda_0 \#\mathcal{M}$, so it suffices to bound $\#\mathcal{M}$. Partition $\mathcal{M}$ by size

$$\mathcal{P}_j = \{T \in \mathcal{M} : 2^{-(j+1)} \leq |T| < 2^{-j}\}, \quad j \geq 0, \quad \#\mathcal{M} = \sum_{j \geq 0} \#\mathcal{P}_j.$$

Proof — Steps 3 & 4

- Step 3 — Bounding $\#\mathcal{P}_j$ (geometry):** Elements of $\mathcal{P}_j$ are **disjoint**: if $T_1, T_2 \in \mathcal{P}_j$ had intersecting interiors then, by the bisection procedure, one contains the other, say $T_1 \subset T_2$, so $|T_1| \leq \frac{1}{2}|T_2|$ — contradicting the definition of $\mathcal{P}_j$. Since each has area $\geq 2^{-(j+1)}$,

$$2^{-(j+1)}\#\mathcal{P}_j \leq |\Omega| \quad \Rightarrow \quad \#\mathcal{P}_j \leq |\Omega| 2^{j+1}. \quad (\text{geom})$$

- Step 4 — Bounding $\#\mathcal{P}_j$ (regularity):** Every $T \in \mathcal{P}_j$ was marked because $e_{\mathcal{T}}(T) > \delta$, and $h_T = |T|^{1/2} \leq 2^{-j/2}$, so

$$\delta < e_{\mathcal{T}}(T) \lesssim h_T^r |u|_{W_p^2(T)} \lesssim 2^{-jr/2} |u|_{W_p^2(T)}.$$

Raising to the p -th power (to account for summability of $|u|_{W_p^2(T)}$) and summing over $T \in \mathcal{P}_j$,

$$\delta^p \#\mathcal{P}_j \lesssim 2^{-jrp/2} |u|_{W_p^2(\Omega)}^p. \quad (\text{sob})$$

Proof — Step 5 (crossover and summation)

- **Crossover index:** Bounds (geom) and (sob) are complementary and cross at an index j_0 with

$$2^{j_0} \approx \delta^{-p} 2^{-j_0 \frac{rp}{2}} |u|_{W_p^2(\Omega)} \Rightarrow 2^{j_0} \approx \delta^{-1} |u|_{W_p^2(\Omega)} |\Omega|^{-1/p},$$

because $1 + \frac{rp}{2} = p$.

- **Summation:** Both geometric series, for $j \leq j_0$ and $j > j_0$, relate to 2^{j_0} as follows:

$$\sum_{j \leq j_0} 2^j \approx 2^{j_0}, \quad \sum_{j > j_0} (2^{-rp/2})^j \approx (2^{-rp/2})^{j_0} = 2^{-(p-1)j_0}.$$

Therefore

$$\#\mathcal{M} \lesssim |\Omega| 2^{j_0} \approx \delta^{-1} |u|_{W_p^2(\Omega)} |\Omega|^{1-1/p}.$$

This is statement (2), and together with (1) completes the proof. □

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- ② **Convergence rate:** $\|\nabla(u - I_{\mathcal{T}} u)\|_{L^2(\Omega)} \lesssim |u|_{W_p^s(\Omega)} N^{-\frac{s-1}{2}}$ is governed by the Sobolev line $\text{sob}(W_p^s) = \text{sob}(H^1)$ (DeVore diagram).
- ③ **Bisection algorithm:** newest-vertex bisection with compatible labeling builds graded, conforming, shape-regular meshes by local operations.
- ④ **Complexity (BDV/Stevenson):** $\#\mathcal{T}_k - \#\mathcal{T}_0 \leq \Lambda_0 \sum_j \#\mathcal{M}_j$ — total marks control total DOFs.
- ⑤ **Greedy optimality:** $u \in W_p^2$ ($p > 1$) exhibits optimal decay rate $N^{-1/2}$, i.e. $u \in \mathcal{A}_{1/2}$ approximation class to be introduced in Lecture 9..

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Next: Lecture 5 — a posteriori error analysis (computable estimators replacing $e_{\mathcal{T}}(T)$).